Nadakkaam

Peer-Led Walking Circles for Middle-Aged Homemaker Mothers in Kochi

Project Proposal: Her Mind, Her Strength

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1. Project Title and Executive Summary

Title: Nadakkaam: Peer-Led Walking Circles to Reduce Isolation and Improve Well-Being Among Middle-Aged Homemaker Mothers in Kochi

Executive summary. Many women aged 40 to 75 in low- and lower-middle-income homes spend almost their whole day indoors, looking after everyone else. They have little time, little space, and often little permission to step out for themselves. Over the years this can bring loneliness, low mood, and neglect of their own health. A preliminary survey of 30 mothers in my own family and neighbourhoods found that most did no regular physical activity beyond cooking and cleaning, and many were distressed at being unable to go out and meet friends.

Nadakkaam ("let's walk" in Malayalam) is a nine-month programme that brings these women together in small walking circles in their own neighbourhoods. Each circle has about 20 women, led by trained peers, walking at a gentle pace three to five times a week. Each woman earns one pair of sneakers after completing 21 days of brisk walking with her circle, so the gift rewards a routine rather than being given before it starts. Once a month, the circle sits together for a short guided conversation on stress, sleep, and health, supported by a clinical psychologist, and women showing serious distress are referred to professional care. The pilot covers five clusters and 100 women, is evaluated with validated questionnaires and interviews, costs about ₹3,700 per woman, and is designed to continue under community leaders after funding ends.

2. Problem Statement and Evidence

2.1 Mental health and gender

Depression and anxiety are leading causes of disability worldwide. A large community survey in India linked women's common mental disorders to gender disadvantage, including low autonomy and restricted movement (Patel et al., 2006). The National Mental Health Survey of India found a wide gap between people with mental health conditions and those receiving treatment (Murthy, 2016). Middle-aged women often carry extra burdens: caring for children and ageing parents, menopause, money worries, and almost no time of their own.

2.2 Low physical activity

Women are less physically active than men worldwide, and insufficient activity is especially common in South Asia (Guthold et al., 2018). The ICMR-INDIAB study found that many Indian adults are inactive, in cities and villages alike (Anjana et al., 2017). Housework is tiring, but it is not the steady, moderate activity that benefits health, and it rarely feels restorative.

2.3 Activity, nature, and mood

Studies following people over time link higher physical activity with a lower risk of later depression (Mammen & Faulkner, 2013), and a meta-analysis found a clear antidepressant effect of exercise in people who are depressed (Schuch et al., 2016). Exercising outdoors in green settings may add mood benefits beyond exercising indoors (Thompson Coon et al., 2011). The World Health Organization recommends at least 150 minutes of moderate activity a week for adults and notes that some activity is better than none (Bull et al., 2020).

2.4 The role of company

A review of 148 studies found that people with strong social relationships had much better odds of survival (Holt-Lunstad et al., 2010). Walking groups specifically have been shown to lower blood pressure, resting heart rate, and body fat, and to improve depression scores (Hanson & Jones, 2015). A group also helps people keep going on days when they would otherwise stay in.

2.5 What mothers in our own community told us

To check whether the published evidence matched what mothers around me experience, I carried out a small, informal preliminary survey. I spoke with 30 women aged 45-75 from my family and neighbourhoods in Kerala, mostly homemakers from low- and middle-income households, asking about their daily routine, how much they move and go out, and how they feel about it. Participation was voluntary, and no names or identifying details were recorded. Two patterns stood out:

- **Little purposeful physical activity.** 27 women said that apart from cooking and cleaning they do no regular activity such as walking or exercise.
- **Distress about being confined at home.** 19 women were unhappy that they could not go out with friends, and many linked this to feeling low, restless, or isolated.

These accounts match the research above. The survey was small and informal and the women were not randomly selected, so it is not representative evidence. I used it to shape the design, not to prove the problem. Its main lesson is that the women themselves named two missing things: a routine of movement and company of their own age. Nadakkaam provides both.

2.6 The gap

Running clubs are now common in Kochi, but they are built mostly for younger, fitter, and wealthier people. A 55-year-old homemaker may feel she will slow everyone down, may not own suitable shoes, or may worry what people will say. Very few programmes are designed around her pace, comfort, and circumstances.

2.7 Where the idea came from

On a recent trip to South Korea and Japan, I saw older people walking together in neighbourhood parks and using outdoor exercise equipment every day as an ordinary habit. It made me ask why our mothers do not have the same. This proposal adapts that idea to Kerala rather than copying it.

3. Target Population

Who they are. Women aged 40 to 75 from low- and lower-middle-income households in selected wards of Kochi, mainly homemakers or in part-time or informal work, who spend most of their day at home.

Pilot size. 100 women in five neighbourhood clusters of about 20 each, reached through Kudumbashree groups, ward councillors, ASHA and Anganwadi workers, residents' associations, and local clinics.

Who can join. Women who live in the cluster area, can walk 15 minutes independently (with rests), and give informed consent. Women with unstable heart conditions, uncontrolled high blood pressure, recent surgery, severe mobility problems, or acute illness will be asked to get medical clearance first. A woman in acute mental health crisis will be connected to professional care first and can join later.

Why this group. They are many, they are overlooked, and their families depend on them. When a mother's well-being improves, her household benefits too.

4. Aim and Objectives

Aim: To improve the mental well-being and social connection of middle-aged homemaker mothers in Kochi through sustainable, peer-led walking circles over nine months.

Objectives:

1. Set up five walking circles with 100 women and train 10 peer walk leaders by Month 2.
2. By Month 6, have at least 60% of participants attending an average of three sessions a week.
3. By Month 9, raise average activity to at least 150 minutes a week (self-reported log).
4. By Month 9, have at least half of participants gain 10 or more percentage points on the WHO-5 Well-Being Index.
5. Reduce average depression (PHQ-9) and loneliness (UCLA 3-item) scores from baseline to endline.

6. Offer referral within one week to every woman who screens positive for significant distress.
7. Hand the circles over to trained community leaders by Month 9.

5. Proposed Intervention

Nadakkaam draws on the Behaviour Change Wheel (Michie et al., 2011), which holds that people change behaviour when they have the capability, opportunity, and motivation. Its two main features, a routine of movement and company of one's own age, come directly from what the surveyed mothers said was missing.

5.1 Outreach talks

Team members give short, friendly talks in Malayalam at community halls, Kudumbashree meetings, and residents' associations. The talks treat walking and time with friends as a normal part of a mother's day and answer the usual worries: "I have no time," "What will people say?" and "I am too old and unfit." Women who already walk share their stories.

5.2 Peer-paced walking circles

Each circle meets three to five times a week for 30 to 45 minutes, in the early morning or late afternoon so it fits around household work, in local parks, school grounds, or quiet streets. The pace is set by the slowest walker, with no targets, rankings, or competitions. Sessions begin with a gentle warm-up and end with stretching and tea or water. Where outdoor gym equipment exists, simple supervised use is added after the first month. In the monsoon, circles move to a community hall for indoor walking and chair-based exercises.

5.3 Sneakers earned after 21 days of walking

Each woman earns one pair of walking sneakers, fitted to her size, after completing 21 days of brisk walking with her circle. Brisk means a steady pace that is brisk for her: she is slightly warmer and breathing a little faster but can still hold a conversation. The 21 days need not be consecutive, so a day missed for illness or a family duty does not reset the count; at three to five sessions a week, most women will reach it in five to seven weeks. Her walk leader keeps a simple attendance log.

The reason for earning rather than receiving is to protect the habit. If sneakers were handed out at enrolment, a woman could take them and stop coming. Tying them to 21 days of walking rewards a routine she has already started. Habit research suggests that walking becomes automatic after about two months on average (Lally et al., 2010), so 21 days is an early commitment milestone and not a guarantee, and the circle and monthly wellness conversations

continue to support her afterwards. Until she earns her sneakers, she walks in her own footwear, and leaders advise on safe options. Where illness or a family emergency interrupts a woman who is clearly committed, the leader may use discretion. No other kit or clothing is provided.

5.4 Monthly wellness circles and referral

Once a month, each circle holds a 45-minute seated conversation led by a trained peer walk leader using a simple guide, with a clinical psychologist or counsellor joining some sessions. Topics include stress and sleep, affordable everyday nutrition, menopause and health, talking with family, and when and where to seek help. The aim is to make it normal to talk about feelings and to connect women with services.

5.5 Peer walk leaders

Two women per cluster (10 in total) are chosen from the participants or neighbourhood as women who are relatively more active, good listeners, and show good team spirit. They can be selected by the team members.

6. Implementation Plan

Phase	Months	Main activities
Preparation	1–2	Ethics approval; partnerships with Kudumbashree, ward offices, a clinic, and a mental health professional; choose five clusters; recruit and train 10 leaders; buy sneakers for the milestone; finalise Malayalam materials and consent forms; map safe routes and indoor alternatives.
Recruitment and baseline	2–3	Community talks; enrolment and consent; health screening camp; baseline questionnaires and survey-style interviews.
Active phase	3–8	Walking circles 3–5 times a week; monthly wellness circles; fortnightly check-ins with leaders; monthly supervision by the mental health professional; attendance tracking; sneakers presented as each woman completes 21 walking days; mid-point review in Month 5 using participant feedback.
Evaluation and handover	8–9	Endline measures; interviews and focus groups; celebration event; handover of guides and contacts; final report shared with local government.

Team: a part-time project coordinator, 10 peer walk leaders, a clinical psychologist or counsellor (consultant), and student volunteers.

7. Innovation and Expected Outcomes

7.1 What is new

- **Designed for the overlooked woman.** Most fitness programmes target younger, wealthier women or run through clinics. Nadakkaam is built for homemaker mothers who fall outside both.

- **Shaped by mothers' own voices.** The design responds to what women in our community said they lacked.
- **Belonging before performance.** Same-age, same-pace groups remove the intimidation that keeps many women away from existing clubs.
- **A walk as the front door to mental health.** Women sign up for a walk and a chat, not for "therapy," which reduces stigma.
- **A reward that builds a habit.** Sneakers are earned after 21 days of walking, so a low-cost gift reinforces the routine instead of replacing the effort.
- **A cross-cultural idea, adapted locally.** East Asian community exercise habits reshaped for Kerala's language, climate, and social setting.

7.2 Expected outcomes

- **For the women:** better well-being, lower depressive symptoms, less loneliness, more physical activity, and more confidence.
- **For the community:** new friendships and support networks, more women visible in public spaces, and trained local leaders.
- **For the health system:** earlier identification and referral of women who need professional help.

8. Evaluation and Measurable Impact

8.1 Design and measures

A mixed-methods, before-and-after evaluation, with questionnaires at baseline (Month 3) and endline (Month 9), process data throughout, and interviews at the end. Baseline interviews will repeat the preliminary survey's questions on daily routine and time spent outside the home, so the pilot builds directly on it.

Domain	Measure	When
Well-being (primary)	WHO-5 Well-Being Index (Topp et al., 2015)	Baseline, Month 6, endline
Depression	PHQ-9 (Kroenke et al., 2001)	Baseline, endline
Loneliness	UCLA 3-item scale (Hughes et al., 2004)	Baseline, endline
Physical activity	Weekly log of minutes walked	Weekly
Physical health	Blood pressure, BMI	Baseline, endline

Questionnaires will be given in Malayalam by trained volunteers, who will read items aloud for women with limited literacy.

8.2 Process and qualitative evaluation

We will track enrolment, retention, attendance, wellness circles held, leader training, referrals made and completed, and satisfaction. At the end, five focus groups (one per cluster) and 10 to 15 individual interviews will explore what changed, what helped or got in the way, and how families responded.

8.3 Targets and analysis

- At least 70% of women retained to endline, and 60% attending an average of three sessions a week.
- At least 70% of enrolled women reaching 21 walking days (earning their sneakers) by Month 5, with continued attendance tracked afterwards.
- At least 50% improving by 10 or more percentage points on the WHO-5 (Topp et al., 2015).
- A meaningful drop in average PHQ-9 and loneliness scores.
- Referral offered within one week to 100% of women who screen positive for significant distress.

We will compare baseline and endline scores, summarise attendance and activity, and analyse interviews by theme. All results will be reported honestly, including any showing no change.

9. Budget

All figures are estimates in Indian rupees and will be replaced with actual quotations.

Item	Calculation	Cost (₹)
Walking sneakers	100 pairs × ₹800	80,000
Part-time project coordinator	₹15,000 × 9 months	1,35,000
Mental health professional	Lump sum	60,000
Health screening camps	100 × ₹200 × 2	40,000
Closing and handover event	Lump sum	25,000
Subtotal		**3,40,000**
Contingency (10%)		34,000
Total		**3,74,000**

Cost per participant: about ₹3,740 for nine months. Sneakers are budgeted for all 100 women; pairs not earned will be kept for new participants.

10. Ethical and Safeguarding Considerations

Ethics approval. We will obtain approval from an institutional ethics committee before recruiting anyone.

Consent. Consent will be explained aloud in Malayalam and recorded in writing. Participation is voluntary, and a woman may leave at any time without penalty and keeps any sneakers she has earned.

Confidentiality. Data will be stored under participant codes in locked or password-protected files, accessible only to the team. Names will not appear in any report. Photographs will be taken only with separate, explicit consent.

Physical safety. Every woman has a short health screening, and those with relevant conditions need medical clearance first. Walks take place in daylight or well-lit areas, in groups, on routes checked in advance. Leaders carry first aid kits and emergency numbers, and sessions are adjusted for heat and rain.

Mental health safety and referral. Walk leaders are not therapists and will not give clinical advice. A woman who scores above the agreed PHQ-9 threshold, or answers positively to the self-harm item, will be contacted by the mental health professional the same day. Referral options include the district mental health programme, government and private clinics, and Tele-MANAS (14416). Anyone at immediate risk will be helped to reach emergency care.

Domestic and family safety. Some women may disclose violence. Leaders are trained to listen without judgement, never to confront the family, to keep confidences unless a life is at risk, and to refer to services such as the Women Helpline (181).

Family and community sensitivity. In some homes a woman's movement outside is a sensitive matter. Outreach will include families, for example through a short information session, and will stress that the programme is safe, respectable, and group-based. No woman should be put in conflict at home because she takes part.

Inclusion and conduct. The programme is free, held in the local language, and welcomes women of all religions, castes, and abilities, with chair-based exercises for those with limited mobility. All team members will complete basic training in research ethics and safeguarding and will never pressure a woman to join or to share personal information.

11. Limitations

- **Informal preliminary survey.** The survey of 30 women was a convenience sample from my own network, with no validated tools, so it informed the design but is not population-level evidence.
- **Small pilot, no control group.** With 100 women and no comparison group, we cannot prove that Nadakkaam caused any change. Results will be treated as promising evidence for a larger controlled study.
- **Self-reported data.** Questionnaires and logs may be affected by social desirability.
- **Limited generalisability.** Five Kochi clusters may not represent other regions.
- **Dropout and short follow-up.** Household duties, illness, and weather may reduce attendance, which flexible timing and indoor options aim to limit. Long-term maintenance will not be measured; a six-month follow-up is recommended if funds allow.
- **Continued participation is not guaranteed.** Earning sneakers after 21 days of walking reduces the risk that women stop once they receive them, but it cannot remove it. A woman may keep attending only until she has her sneakers, and 21 days is shorter than the time habits usually take to form (Lally et al., 2010). Peer encouragement, flexible timing, and the monthly wellness circles aim to sustain participation, and attendance and dropout after the milestone will be tracked and reported honestly.
- **Not a treatment.** Nadakkaam is supportive and preventive and does not replace professional care for moderate to severe mental illness.

12. Sustainability

- **Community ownership.** Trained leaders, who live in the neighbourhood, continue the circles after funding ends.
- **Low running costs.** Once sneakers are earned and distributed, costs are mainly small leader support and occasional refreshments.
- **Partnerships.** Kudumbashree networks, ward councils, and health centres can host circles and fold the model into existing women's and health programmes.
- **A replicable guide.** A short Malayalam and English Nadakkaam Starter Guide will explain how to start a circle, run a wellness conversation, and use the referral pathway.

- **Funding and advocacy.** Local businesses, corporate social responsibility funds, and local government can sponsor sneakers for new groups, and findings will be shared to push for safe walking paths, outdoor exercise equipment, and women-friendly park hours.

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